

Rothamsted Take-all Wheat Experiment: Data Cleaning and Preliminary Analysis

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Introduction

Take-all is a disease affecting wheat crops caused by the fungus *Gaeumannomyces tritici*. Wheat infected with take-all is severely stunted in growth and grain production. The fungus spreads through the soil and different cultivars of wheat and barley are susceptible, especially second and third wheat crops within the same plots¹. First-wheat experiments were conducted at Rothamsted Agricultural Research Station nearly every year between 2004 and 2023. In total, fifteen years of data were collected along with rainfall, temperature, radiation hours, and relative humidity metrics measured from the Rothamsted weather research station. It is largely found that, of the weather factors analyzed, none provide a significant explanation for the levels of take-all in first-year wheat. The most closely correlated variable to take-all disease in wheat is average rainfall over a particular set of months, namely May, June, and July. However, even these metrics fail to satisfactorily describe the variation in take-all between years.

Additionally, long-term wheat experiments were conducted at Rothamsted between 2015 and 2021 to explore the relationship between take-all disease, plant height, grain yield, and straw yield between five popular varieties of wheat (Hereward, Cadenza, Xi19, Hereford, Duxford). Preliminary analysis shows a strong connection between straw dry matter and grain dry matter, as well as an overall decline in take-all after year four. Moreover, we note that the variate of wheat affects take-all disease in first wheat the most, with the effects of variate on take-all diminishing as years progress.

Lastly, data cleaning for Rothamsted Research is performed to organize, standardize, and format experimental data from various studies at Rothamsted. See Appendix 1 for a step by step guide on how to use the code to clean the provided data.



1 Motivation

Rothamsted conducts research on a variety of agricultural topics, with the ultimate goal of gaining knowledge to produce enough food for a growing population while caring for the environment. They are the longest-running agricultural research institution in the world, with one experiment even being started in 1843 and continuing to this day.

Take-all (*Gaeumannomyces tritici*) is a soil-borne fungal pathogen which invades and destroys root vascular tissue. It is the most important root disease of wheat to study because of its negative impacts on wheat yield². Annual losses in 2nd wheats can reach up to 60% loss in severe disease years³. The common name for the disease is take-all because it “takes all” of the grain from the wheat, which is the part of the plant that people harvest and use to feed the growing population. Yield is predicted to decrease more in the coming years due to climate change³. Factors known to impact take-all build-up include weather, soil type, crop management (e.g. break crops), and wheat cultivar⁴. Wheat cultivar refers to the different varieties of wheat, which can have different traits such as disease resistance and grain quality. It

is important to control take-all build-up (TAB) in the first wheat crop because take-all can heavily impact the subsequent wheat crops. There is a phenomenon called take-all decline, which refers to when take-all begins to decrease after about the fourth wheat crop⁵. However, farmers would have significant loss in their second to fourth wheat crops in order to reach take-all decline, meaning that it is not sustainable to wait for take-all decline. To measure take-all build-up, Rothamsted uses soil core bioassay techniques and collects a variety of information, including the percentage of infected plants and the percentage of take-all infected roots.

1.1 Previous research

Rothamsted has been working on researching take-all disease for many years, and has published several articles describing their findings. In 2010, research on differences in wheat cultivars was published. They discovered that wheat cultivars differ in their ability to build up inoculum of take-all in first wheat crops⁶. In 2018, they published their findings that there is a substantial reduction in take-all disease and grain yield increases when a low take-all inoculum building wheat cultivar is grown in the first year of wheat cropping⁴.

1.2 Research questions

The main objective of this project is to analyze the factors affecting take-all inoculum buildup and severity, along with wheat straw/grain yields. This paper explores the effects of weather factors, including temperature, humidity, sunshine and rainfall, on first-year wheat take-all infection. Beyond weather, the analysis explores the association between straw yield, grain yield, plant height, and take-all buildup during long term wheat experiments. Finally, this project examines the impact of controllable factors such as wheat cultivars on the buildup, decline, and severity of take-all infection. Because different wheat cultivars differ in their contribution to take-all buildup, the study aims to determine which cultivars most effectively mitigate take-all infections and how cultivars relate to take-all decline.

2 Data

All wheat and take-all data is provided by Rothamsted Research Institute, based in the United Kingdom. The WGIN Diversity Trials were conducted from 2004 to 2023 and contain annual samples from several wheat variate harvests. Critically, the percent of wheat crop roots infected and the percent of wheat plants from the sample with take-all present are both recorded yearly. This data is combined with weather data during these years to explore the impact of any combination of the four weather variables explored, namely, rainfall, temperature, solar radiation, and relative humidity on TAB. Weather data is provided by the Rothamsted Meteorological Station. Data is missing from 2008, 2012, 2018, 2020, 2022 for a variety of reasons, but mostly due to poor weather conditions limiting harvesting and sampling. Furthermore, data regarding infected root percentages and plant percentages for first-wheat diversity trials for 2012, 2018, and 2022 is supplemented with estimates from biologists from Rothamsted Research to reflect estimated take-all levels in years lacking sufficient records.

Moreover, long term trials (exp. 760, 773, 778, and 789) with 5 different wheat varieties, Hereford, Hereward, Cadenza, Xi19, and Duxford were conducted between 2015 and 2021. The response variables of interest from this study are plant height (cm), take-all (i.e. metrics used in the first-wheat studies), and several measurements of grain yield. For example, winter wheat dry matter was recorded to quantify plant health and growth and describes the percentage of dry mass in the wheat sample. The goal

of this second analysis is to examine the correlation between take-all buildup (TAB), plant height, and grain yield.

3 Data cleaning

Rothamsted Research uses the MIAPPE standard (Minimum Information About a Plant Phenotyping Experiment), which was designed to standardize data from plant phenotyping experiments, but could be used to format the data for this take-all experiment. The Excel files have the following worksheets:

- Study: the metadata for the field trials or soil core bioassays
 - Low Tab Trait Pedigree Survey
 - Diploid Wheat and Inoculum Build-Up
- Data file: describes the data and includes links to all the data files
- Biological material: explains the biological materials used, such as the wheat varieties
- Experimental Factor and Experimental Factor Level: gives details about the experiment, like the varieties used and the rates of nitrogen application
- Variable Dictionary: gives definitions for all the variable names

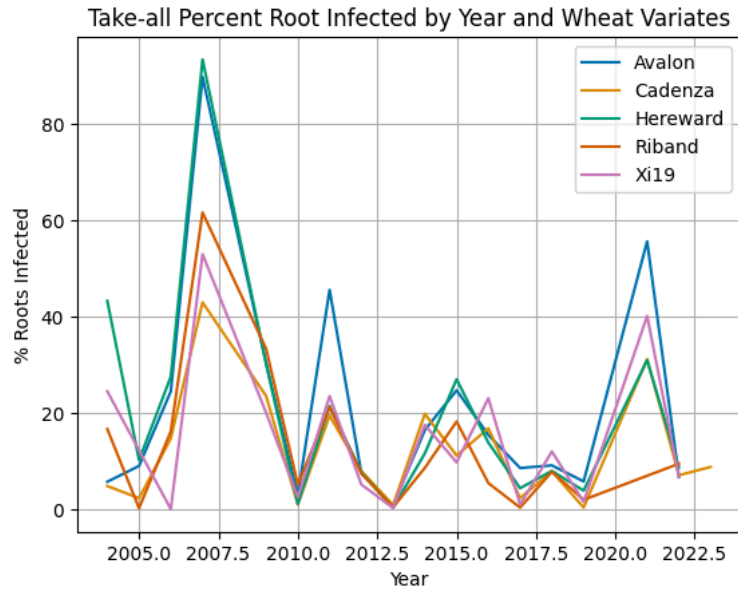
The data is compiled from different sources over a 20 year period of time and therefore uses inconsistent formatting and variable names. While many of the files were already cleaned by the data scientist at Rothamsted, many more still needed to be cleaned in order to standardize the data and remove any inconsistencies.

4 Methods

For both studies, data is stored in separate files and organized by year. Univariate analysis is performed to understand the distribution of variables and their properties. For example, in the first-wheat study, the range of TAB over successive years is first explored without relation to weather. Next, correlation is explored using a variety of methods including crosstabs, heatmaps, and pairplots. When explanatory and response variables present an above-average correlation, several techniques are then implemented in order to model response variables from explanatory. For example, rainfall is shown to be most prevalent to first-wheat TAB during particular months of the year (see 5.1.1 below), so a multivariate regression analysis is performed to model TAB using rainfall from these months. Lastly, a post-hoc analysis is performed to test for statistical significance and ensure that outliers and probabilistic distributions are accounted for when modeling. For example, it is noted that one first-wheat year, namely 2007, has unusually high TAB levels, which must be accounted for when modeling.

5 Results

It is clear that take-all buildup varies between years of first-wheat harvests. It is also known that different varieties of wheat respond differently to take-all buildup in soil, specifically when analyzing root infection rates (see section 1.1). Below, Figure 1 displays the average root infection rate across the five most planted first-wheat varieties between 2004 and 2022. We note that while more susceptible varieties, such



as Avalon, have consistently higher rates of root infection, root infection rates for all variates seem to increase and decrease in unison.

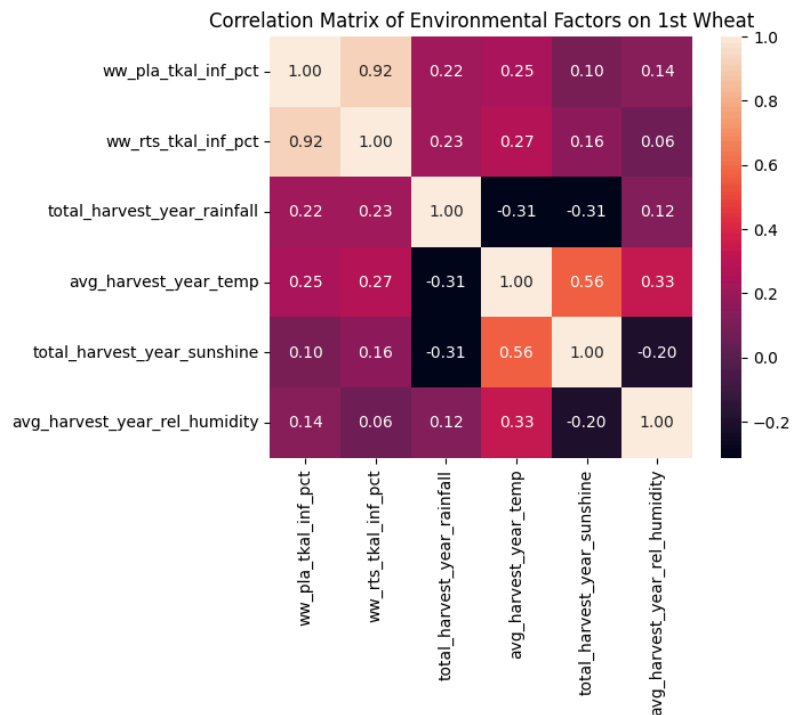
Figure 1 (left): Take-all root infection rates per variate (5 most harvested) of wheat between 2004 and 2022.

Figure 2 (below): Heat map displaying the correlations between different environmental factors and take-all.

It is suspected that some external factor(s) influence all first-wheat take-all buildup and decline. Thus, a variety of weather factors are analyzed in search of strong correlations and effect models.

5.1 Weather

Environmental factors, including rainfall, temperature, sunshine, and humidity, are suspected to impact the plant take-all infection percentage and roots take-all infection percentage on first-year wheat. Figure 2 (right) provides an overview of the correlations between harvest year environmental factor summaries and take-all infection percentages. The heatmap shows that none of the yearly summaries of environmental factors had a strong correlation to take-all infection. This lack of strong correlation prompted a deeper



investigation into the impacts of weather during specific months of the harvest year on take-all infection percentages.

Figure 3 (below) presents the monthly correlations between weather factors and the annual percentage of root take-all infection averaged during 2004 to 2023.

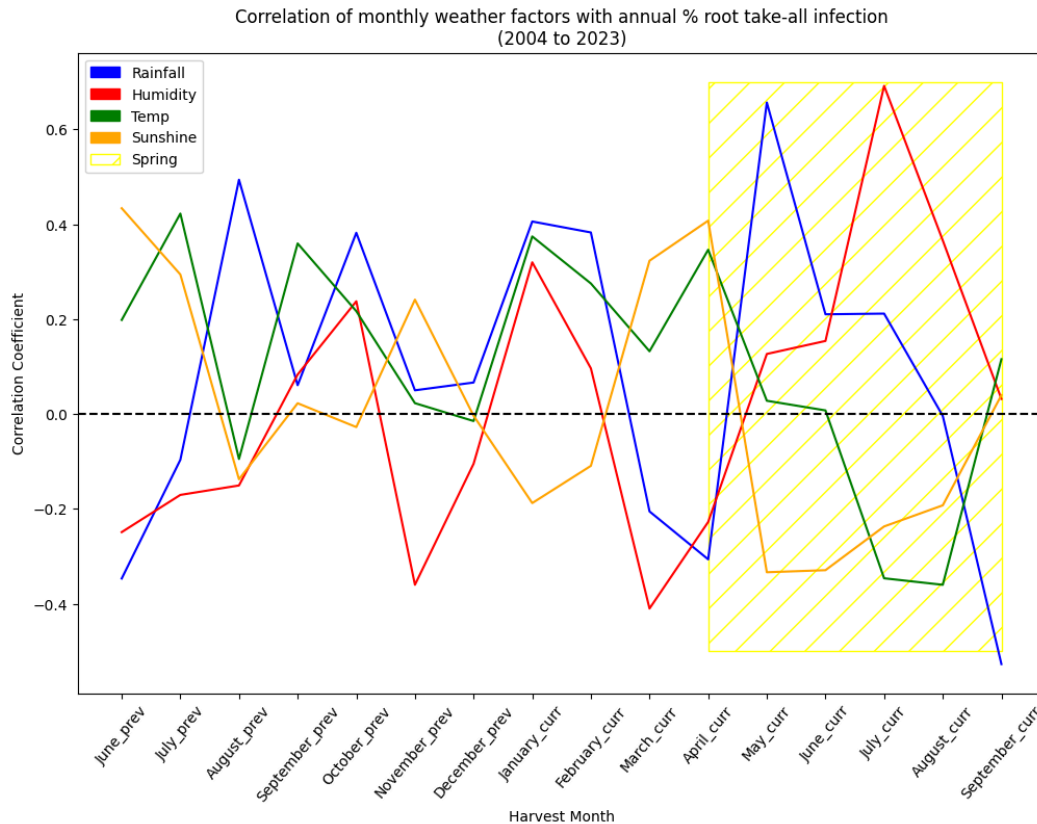


Figure 3: Correlation of monthly weather factors to annual percent root take-all infection

Generally, the correlations of rainfall and humidity follow a similar pattern. In addition, the correlations of temperature and sunshine follow a similar pattern. From investigation, weather from the current April to harvest appear to have the highest effects on take-all root infections. The line plot highlights that during these months, temperature and humidity positively correlate with root take-all infection percentages. This correlation could be related to increased moisture from rain and humidity causing the fungus to grow significantly. From the current April until harvest, temperature and sunshine are negatively correlated with root take-all infection percentage, suggesting that the higher temperatures and increased hours of sunshine correspond with lower levels of take-all buildup. We suspect more sunshine and higher temperatures could dry out the soil, causing slower growth of fungus.

To build on the previous visualization, the cumulative absolute values of correlations between monthly weather factors (including cumulative hours of sunshine, total rainfall, average temperature, and average humidity) and root take-all percentages are shown in the stacked bar chart below (Figure 4). The stacked bar chart highlights the combined effects of environmental variables during each month of the harvest year. In particular, the previous and current July have the largest cumulative absolute value of

correlations, emphasizing that weather during those months of the harvest year has the strongest impact on the take-all root infection percentage.

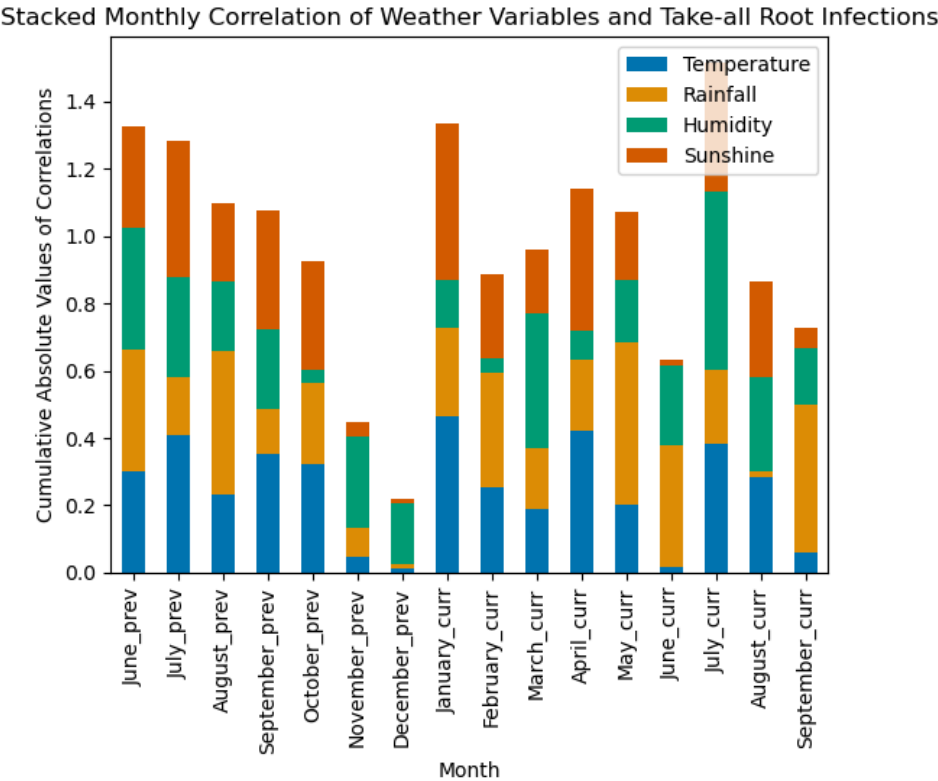


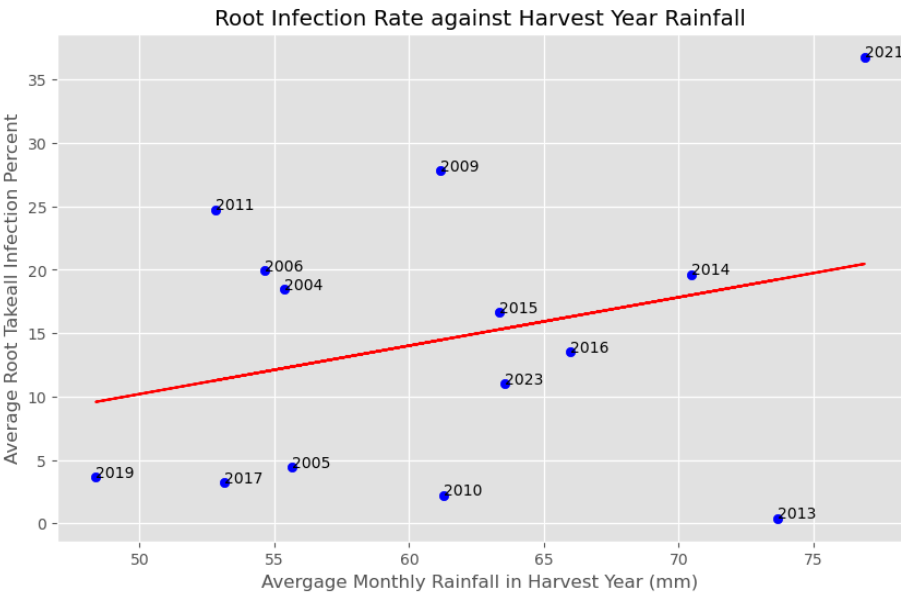
Figure 4: Cumulative correlations of four weather factors to take-all root infection by month

Overall, yearly weather summaries have minimal impacts on the root take-all infection percentages. Figure 4 shows the variations in correlation of weather factors and root take-all infection percentage based on month of the harvest year and reveals that certain variables have stronger correlations during different months. These results initiate further analysis of the impact of individual weather factors on the root take-all percentages.

Figure 5: Linear regression of monthly rainfall to avg. root infection

5.1.1 Rainfall

To investigate rainfall’s influence on take-all root infection percentages, a linear regression is performed utilizing average monthly rainfall versus average root take-all infection percentage and the graph of the regression is shown in Figure 5. The R-squared value resulting from this linear regression is 8.6% and the p-value of 0.308 for the



slope, indicating the insignificant effect total harvest year rainfall has on take-all root infection percentages.

Based on the previous investigation of monthly weather impacts on take-all root infection, Figure 6 is created with groupings of months to reveal that the rainfall during the current May, June, and July has a moderate correlation to the plant take-all root percentage. To further research this correlation, a multivariate regression is run to identify the impact of the rainfall during May and June on the root take-all infection percentage. The resulting R^2 is 0.345, showing a stronger association between May and June rainfall and root take-all infection percentage than total harvest year rainfall and root take-all infection percentage.

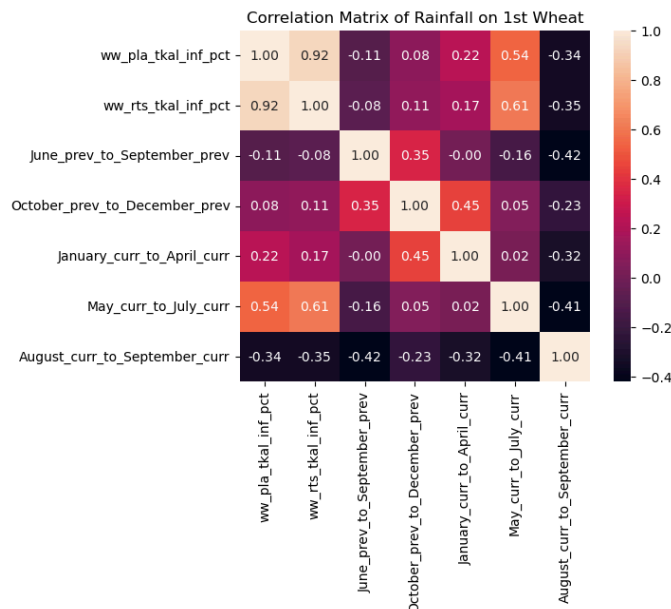


Figure 6: Correlation matrix of rainfall (aggregated in 3 month intervals) with take-all infection

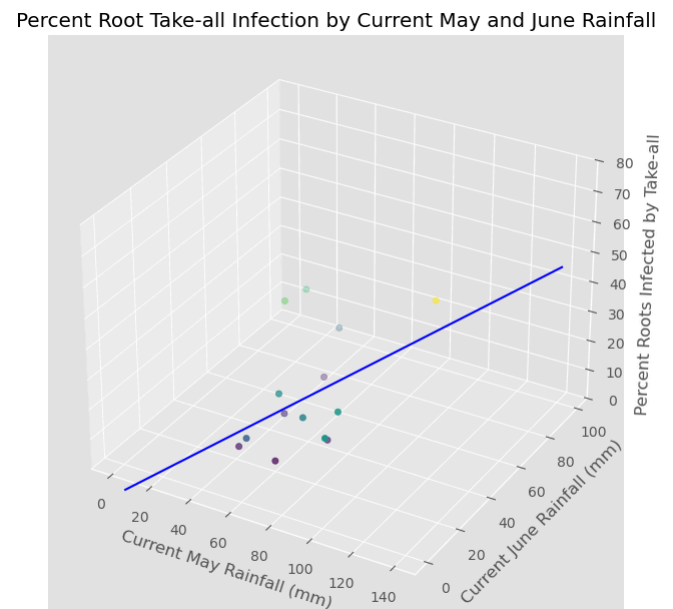


Figure 7: 3d plot with multivariate regression for May/June rainfall and take-all

Overall, there is not a strong correlation between harvest year rainfall and root take-all infection percentage. Moderate correlation is evident between current May and June rainfall and root take-all infection percentage.

5.1.2 Remaining weather variables

A similar process to the investigation with rainfall is repeated with the following variables: days without rain, temperature, humidity, and sunshine. These investigations concluded with no significant results.

5.1.3 Modeling take-all percentage with yearly weather data

An initial linear regression model is created using predictor variables of wheat variety, harvest year, and average temperature, total rainfall, total sunshine, and average humidity in that harvest year. The model aims to show which of these weather factors have significant effects on root take-all percentage.

```
(Intercept)      -73.39509    65.57951   -1.119  0.264004
harvest_year      -0.76492     0.32193   -2.376  0.018160 *
avg_harvest_year_temp  19.66742    4.14369    4.746  3.28e-06 ***
total_harvest_year_rainfall  0.03936    0.01051    3.745  0.000218 ***
total_harvest_year_sunshine -0.02461    0.01366   -1.801  0.072693 .
avg_harvest_year_rel_humidity -1.45076    0.99204   -1.462  0.144729
variety_fAvalon    11.73131    3.67821    3.189  0.001584 **
variety_fHereward   7.02708    3.67821    1.910  0.057074 .
variety_fRiband     2.20563    3.73712    0.590  0.555526
variety_fXi19       3.59658    3.72587    0.965  0.335210
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Figure 8: Linear regression model with several variables on root take-all infection

We find that harvest year, temperature and rainfall have significant effects. Harvest year has a negative effect, that is, as time goes on, there is expected to be less take-all infection. Average temperature and total rainfall have a positive effect, so in years with more rainfall and higher temperature, there is expected to be more take-all infection. This model has $R^2 = 0.185$, so it does not come close to fully explaining the variation in take-all percentage. The varieties also have an effect, but only Avalon, and marginally Hereward, have any effect that is significantly different from Cadenza.

5.1.4 Modeling take-all percentage with monthly weather data

Several variable selection procedures are performed in order to determine which monthly weather variables should be included in a model. A subset regression analysis is performed to find an optimal model for every size subset. This is performed for each of temperature, rainfall, and sunshine in order to determine which months work together to best explain take-all percentage. From this analysis, the most impactful months are found to generally be the months closest to harvest (May, June, July, August) with the exception of January temperature data, which is found to be one of the strongest months for explaining the effect of temperature.

Next, a mixed-effects linear model is created, using the strongest three months of each weather type as the fixed effects, and wheat variety and block as the random effects. This model is made without the data from 2009 since the experiment from that year has a different block structure which includes nitrogen levels as treatments. It also includes only the top 5 most commonly used wheat varieties (Avalon, Cadenza, Hereward, Riband, and Xi19). While this model finds several predictors to be statistically significant, the R^2 value only reaches 0.573. Using this, a similar model is created using only summer rain (May + June + July), January temperature, and August sunshine as fixed effects.

```
Fixed effects:
              Estimate Std. Error    df t value Pr(>|t|)
(Intercept)   8.73836    7.36614 164.14103   1.186   0.237
summer_rain    0.18084    0.02756 271.41675   6.562 2.67e-10 ***
January_temp   4.74366    1.00065 271.20725   4.741 3.45e-06 ***
August_sun    -0.22975    0.04385 271.21410  -5.239 3.24e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Figure 9: Mixed-effects linear model with three strongest months and their weather factors

This model finds all three predictors to be strongly significant, with rain and temperature having a positive effect, and sun having a negative effect, however the predictors do not explain all variation in the model, since the R^2 value was 0.476.

After this, a Lasso (Least Absolute Shrinkage and Selection Operator) regression is performed on the data from every month. This algorithm shrinks the majority of coefficients to zero, showing that they are not strong enough predictors of take-all. The ten selected months are input into a mixed-effects model with similar structure to the previous one, where seven of them are found to be significant, namely May and June rain, January, March, and April temperature, and July, and August sun. This model still only reaches $R^2 = 0.564$, so it cannot explain all variation.

Both approaches agree on May rain and August sun being the monthly weather measures that most strongly predict root take-all percentage. They also find January temperature to be one of the most impactful months for temperature, although temperature seems to have a weaker effect on take-all growth.

5.2 Long-term plant height, straw, grain yield, and TAB

5.2.1 Straw yield and plant height

As mentioned above, waiting for take-all decline is an impractical method of farming. While the percentage of infected plants does decrease over time, it does not happen without substantial losses in crop yields. Figures 10 and 11 show the percent straw yield and heights of each successive crop harvest, split by each wheat variety. For each wheat variety, the yield and height generally decrease from 2015 to 2020 since the break crop.

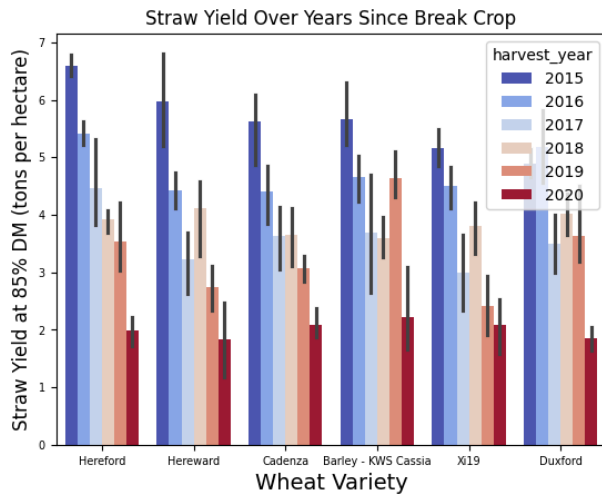


Figure 10: Straw yield by variate over long-term study

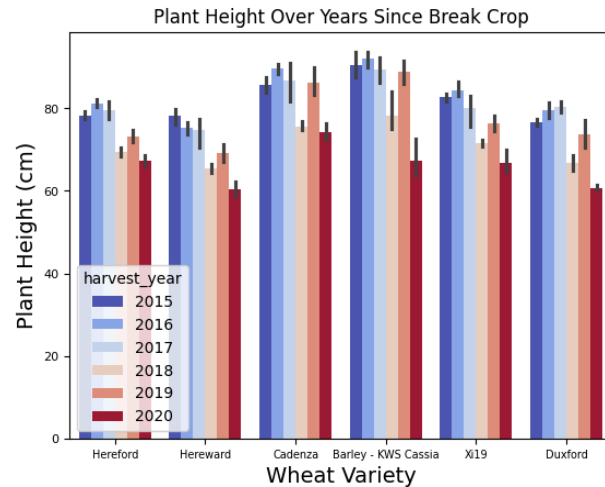


Figure 11: Plant height by variate over long-term study

Furthermore, we find that taller plant heights are associated with a higher straw yield at 85% dry matter. The plant height in centimeters accounts for roughly 28% of the variance in straw yield across plots from all years in the 760 study. We expect that taller plants will have higher mass in straw than shorter plants. However, the relationship is not as significant as first thought. This may be a result of different wheat varieties reacting variably to take-all disease.

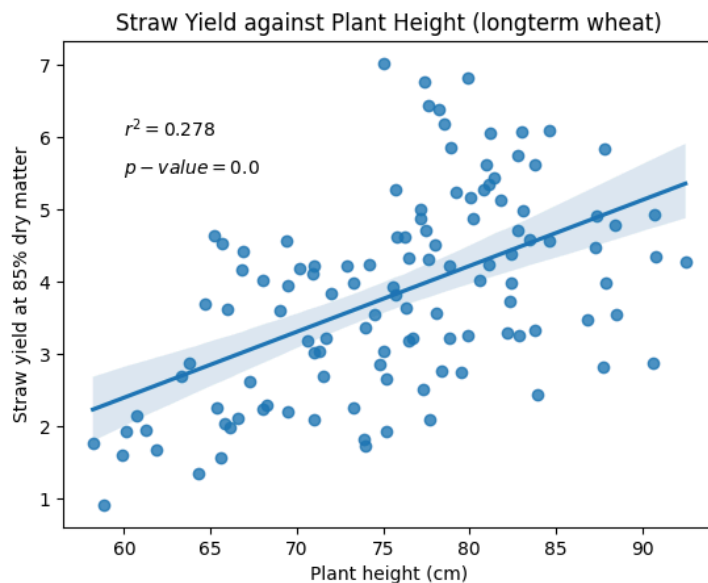


Figure 12: straw yield (tons per hectare) against plant height (cm)

Pearson Correlation Coefficient = 0.527

However, the weak R-squared value between plant height and straw yield in Figure 12 may suggest that other factors are heavily influencing the straw yield.

When broken down by wheat variety, a relationship between plant height and straw yield becomes more apparent. Notably, plant height explained Hereford and Hereward straw yield variance with R-Squared values of 0.587 and 0.501, respectively.

5.2.2 Straw yield and grain yield

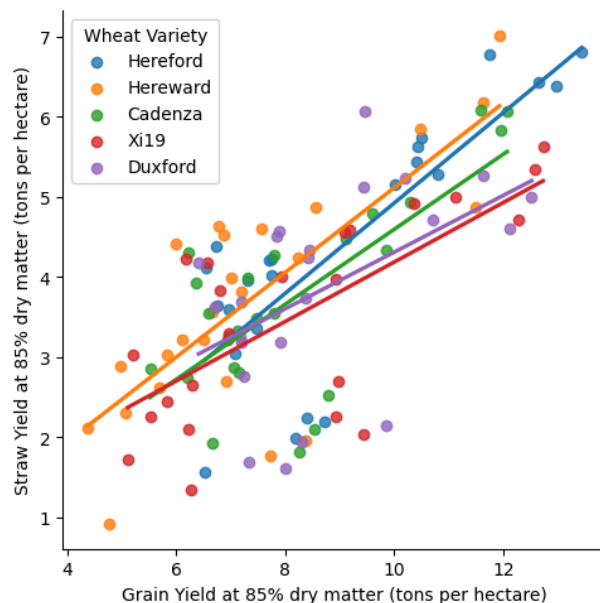


Figure 13: Straw yield against grain yield by wheat variety (both in tons per hectare)

As expected, straw yield and grain yield are closely related. Long-term wheat with higher straw yields are associated with higher grain yields. This suggests that grain yield and straw yield could be used together to form a reliable metric for the health of the plant. Figure 13 plots the grain yield against straw

yield, for each wheat variety. Similar to straw yield against plant height, Hereford and Hereward have the highest R-Squared values of 0.647 and 0.589 respectively. Overall, while the correlations are not very high, we still see the positive relationship between straw and grain yield.

5.3 Plant height and take-all disease

5.3.1 Lack of strong correlation between height and severe take-all

Most surprisingly, there is very little measurable effect of take-all disease on plant height. Contrary to prior belief, the percent of roots infected with take-all within a plot was not, in general, associated with shorter plants. Although it is widely accepted that take-all disease stunts wheat growth¹, the long-term wheat data does not support this conclusion. In fact, the opposite is observed with certain wheat varieties, such as Duxford, where statistical tests suggest that plots with higher levels of take-all disease contained taller plants. In Figure 14 below, plant height is plotted against the percent of plants with severe take-all, with data points grouped by variety factor level. Barley and the Cadenza variate display little to no connection between severe take-all and plant height, possibly explained by their known resistance to the disease. However, the other varieties (Hereford, Xi19, Hereward, and Duxford) of wheat have proportional, statistically significant non-zero sloped relationships (p-values of 0.001, 0.004, 0.166, and 0.000, resp.). We note that all varieties have a proportional relationship rather than an expected inversely proportional one.

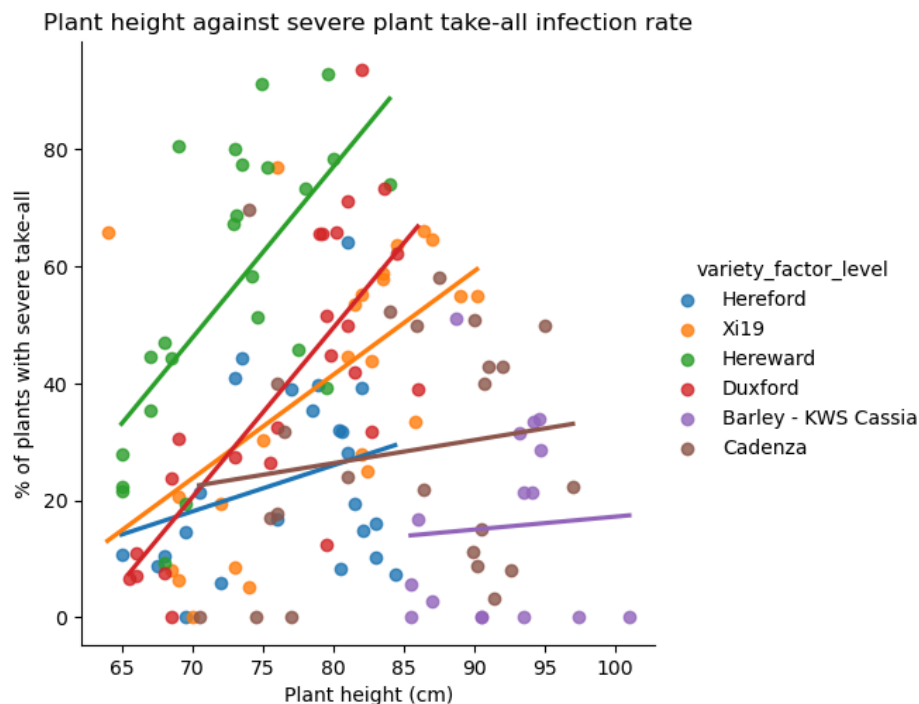


Figure 14: Severe take-all percentage against plant height by variate

5.3.2 Explanations of findings

It is suspected that the findings above are a result of poor measurement strategies. Within plots, 5 soil cores are sampled to measure TAB in the soil. The percent of plants with severe take-all (classified by Rothamsted Research) is then recorded from these samples per plot. Meanwhile, plant height is measured from these samples as well and averaged. It is likely that relevant data points are lost when aggregation

occurs. For example, while the severe take-all percentage metric maintains information regarding the amount of TAB in each plot, the plant height average may not account for the shift in distribution of heights within the plot. It is known that take-all disease is patchy within plots, so the mean height may not be the most statistically significant measurement to record. Skew, kurtosis, other statistics, or even raw data might show more significant findings that reflect the shift in distribution of plant heights as take-all increases within plots.

There may also be biological and ecological explanations for the unintuitive findings above. For instance, perhaps wheat around the patches of take-all is under less stress and is more capable of outcompeting its neighbors, thus increasing its height and obscuring the shorter plants within the data. Boundary plants around the perimeter of each plot are known to grow taller since they have less competition from plants around them, so it is speculated that the boundary plants around the take-all patches may be sampled and weighted against shorter plants.

Additionally, variation of plant height and take-all between harvest years accounts for a large amount of error in modeling. Over time (and studies), the amount of take-all present as well as the weather, sowing methods, nitrogen levels, and countless other variables may have differing effects on plant height, all of which confound the data and create obstacles in analysis.

5.4 Variations over years of consecutive wheat (exp. 760)

5.4.1 Take-all disease

Take-all buildup over the six years of experiment (exp.) 760 sees a general increase in disease during the initial years (2015 to 2018), followed by a period of take-all decline from 2019 to 2020. In 2015, the first year of the study, the percent of roots infected with take-all disease is relatively low as expected, with an estimated population mean between 30% and 40% (95% confidence). The middle years of the study (2016 to 2018) all have estimated population root infection rate means near 60% with 95% confidence, suggesting that take-all thrives in the 2nd, 3rd, and 4th year wheat generations. Finally, the last two years of the study see a decline in TAB, with mean root infection rates estimated at about 40% and 30%,

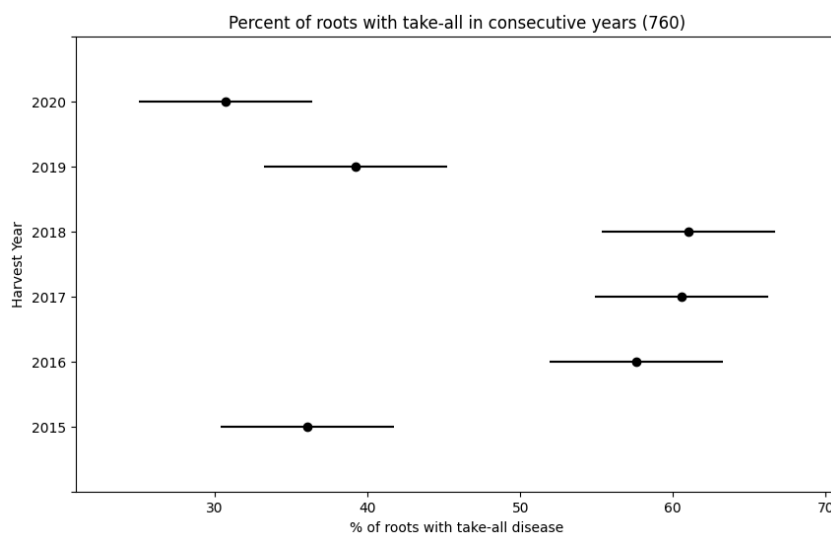


Figure 15: 95% confidence intervals of true mean root infection rate per year

respectively with 95% confidence. Figure 15 displays the harvest year on the vertical axis (reading upwards), and the estimated true percent of roots infected with take-all on the horizontal axis. Standard error bars are provided to represent a 95% confidence interval.

A further breakdown of the take-all buildup and decline per variate of wheat is provided in section 5.5.

5.4.2 Straw yields

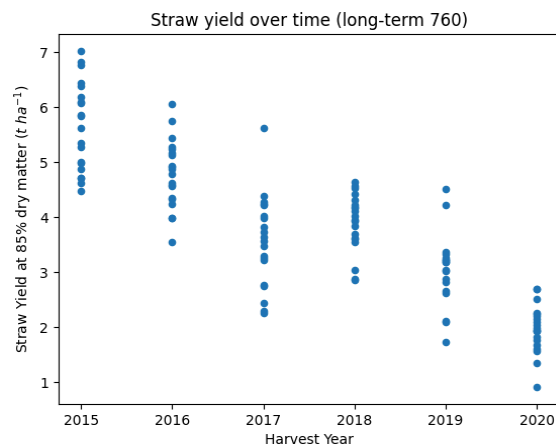


Figure 16: Straw yield over consecutive years of wheat.

Within the long-term study (exp. 760), we see a general decrease in straw yield between 2015 and 2020. When considering all variates together, the straw yield in tons per hectare decreases from an average of around 5.7 tons per hectare in 2015 to around 2.0 tons per hectare in 2020. Additionally, the straw yield population confidence intervals at an alpha of 0.05 for each year are all statistically different from each other (see Figure

17), with the exception of 2017. This suggests that true mean straw yield for each year is decreasing over the entirety of the experiment. However, as seen in section 5.4.1, the take-all root infection percentage did not decrease in the same capacity as straw yield. It follows that some factor(s) are strongly affecting the straw yield throughout the long-term experiment besides the amount of take-all in the soil. It is expected that as take-all grows in the middle years of the study, namely from 2016 to 2018, that the straw yield will decrease. However, the straw yield never recovers to its original levels after take-all decline occurs in the later years (2019 to 2020).

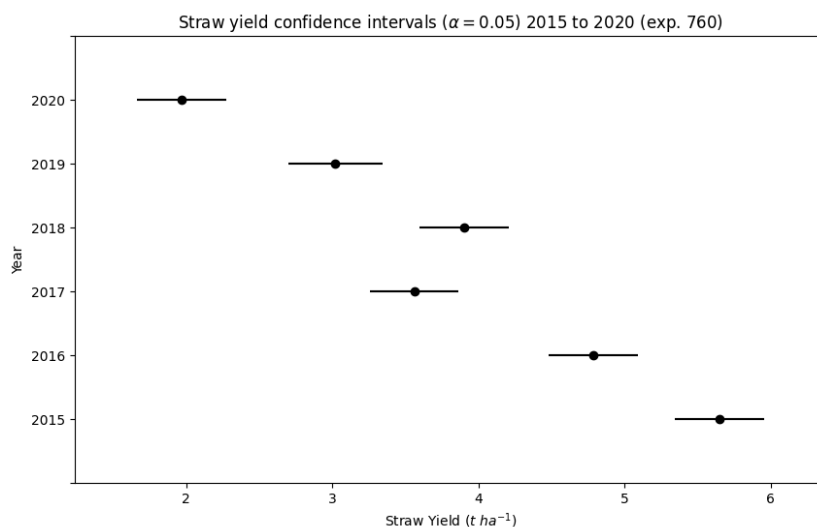


Figure 17: 95% confidence intervals for mean straw yield by harvest year

5.4.3 Grain yields

Interestingly, the grain yield of wheat crop in the long-term experiment follows a similar pattern to the take-all buildup. Grain yield is highest during the first year of the study, then steadily decreases through the middle years of the experiment, then increases again in the last two years. This follows with the take-all buildup and decline periods from section 5.4.1.

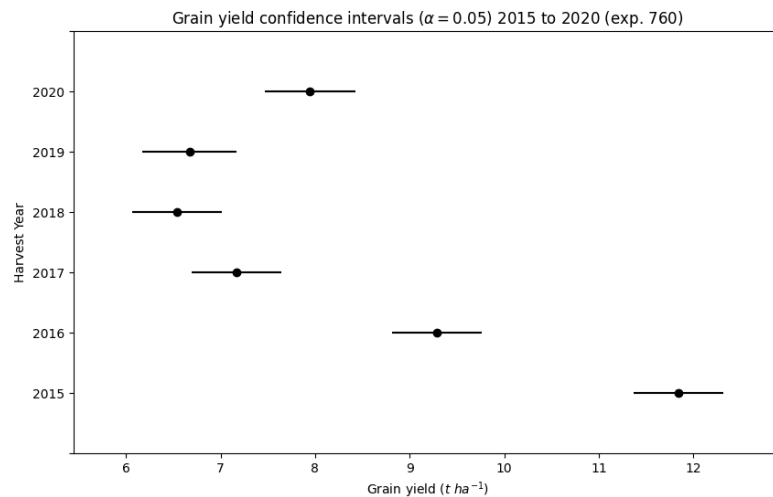


Figure 18: 95% confidence intervals for mean grain yield by harvest year

In 2015, the take-all buildup was relatively low and the grain yield was at its maximum for the study. During the middle years of the study, grain yield decreased dramatically to a low of between 6 and 7 tons per hectare in 2018. Meanwhile, the percent of roots infected with take-all increased to a maximum in 2018 to roughly 62%. We find a correlation coefficient between grain yield and root infection percentage of -0.358.

5.4.4 Plant height

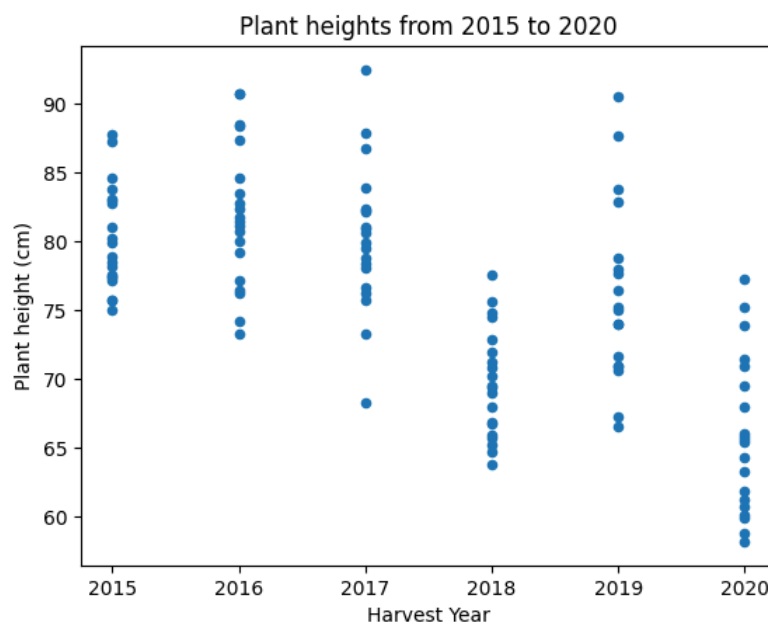


Figure 19: Plant heights for consecutive years of wheat

Following the same trend as straw yield, the plant heights in each plot decrease on average as years progress. Specifically during the height of take-all in 2018, the average plant heights (aggregated from 5 samples per plot) are significantly lower than in the first year, 2015 (p-value of 0.000). Also similarly to straw yield, the plant heights do not seem to recover after

take-all decline occurs, with heights varying in 2019 and dropping again in 2020. This is expected since plant height and straw yield were associated with a pearson correlation coefficient of 0.527 (see section 5.2.1).

5.5 Wheat variety

5.5.1 Take-all buildup in various first wheat cultivars

Based on previous analysis performed by Rothamsted Research, it is determined that wheat cultivars differ in their ability to build up take-all inoculum. Subsequently, Rothamsted Research conducted a series of diversity trials in which they grew 68 different cultivars as first wheats between 2004 and 2023, taking soil core bioassays to collect data on the build up of take-all in the soil for each cultivar. This allows for a comparison of a wide variety of cultivars to those previously determined to have low TAB trait, a low ability to build up take-all in soil, or high TAB trait, a high ability to build up take-all in soil.

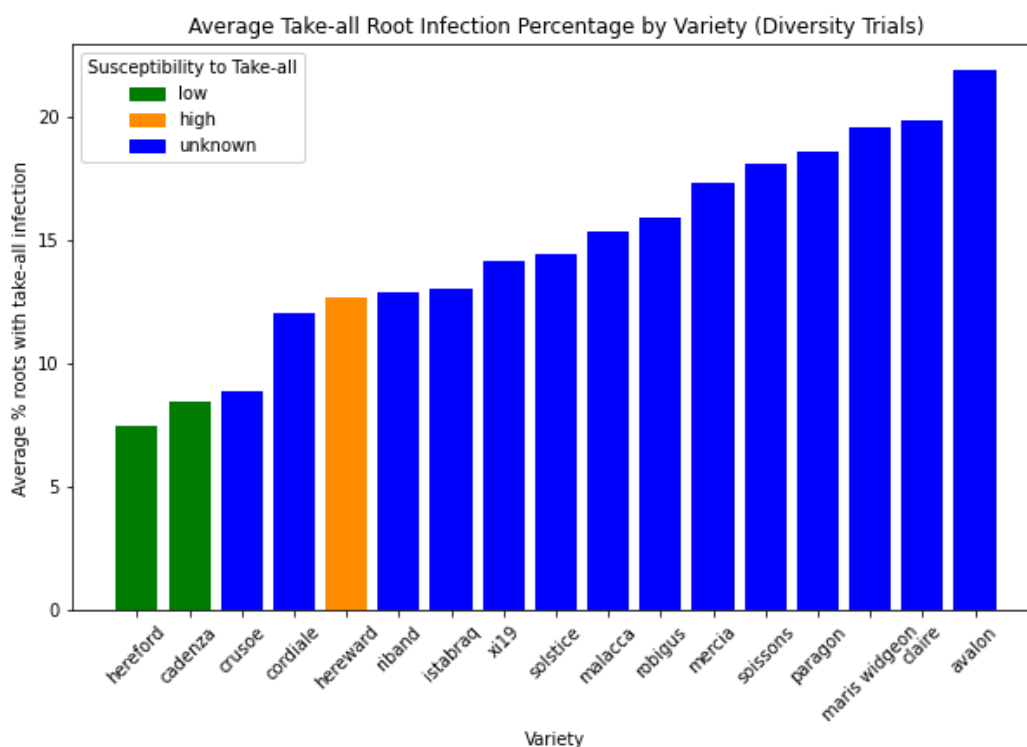


Figure 20: Take-all root infection by wheat variate, ordered ascendingly and colored by known susceptibility

Figure 20 shows the average percent of roots with take-all infection as determined by the soil core bioassay technique for all varieties that were planted in more than 20 plots throughout the course of the diversity trials. Three of the wheat varieties were previously determined to have low TAB or high TAB traits, and are colored accordingly in the figure above. Hereford and Cadenza were previously accepted as having low TAB traits (colored green above), while Hereward is known to have high TAB traits (colored orange above). This allows a basis for comparison for the other varieties whose TAB traits are unknown.

No cultivars are determined to have a lower average percent of roots with take-all infection than those already known to have low TAB traits (Hereford and Cadenza). Only two of the examined varieties, Crusoe and Cordiale, are found to have an average percent of roots infected with take-all between the two

known low TAB varieties and the one known high TAB variety. All other cultivars are found to have a higher average percent of roots with take-all infection than the known high TAB trait variety, Hereward, suggesting a high TAB trait for these cultivars as well.

5.5.2 Impact of low TAB and high TAB first wheat cultivars on take-all in second wheats

To investigate the impact of low TAB trait and high TAB trait cultivars grown as first wheats on take-all disease in a variety of second wheat cultivars, Rothamsted Research performed a series of four two-year experiments beginning in successive years between 2009 and 2012. In the first year of each experiment, four replicates of Cadenza, a low TAB cultivar, and Hereward, a high TAB cultivar, were sown in large plots. In the second year, each of these plots were divided into eight smaller plots and each was drilled with one of eight winter wheat cultivars. During the second year of each experiment, soil core bioassays were once again taken to collect data on the build up of take-all in the soil for each cultivar. The average percent of roots with take-all infection was determined using the soil core bioassay technique. Furthermore, the severity of the take-all disease was recorded through a weighted five category winter wheat take-all index.

Figures 21 and 22 are created to visualize the impact of low TAB and high TAB cultivars as first wheats on the buildup of take-all in the soil during the second wheat harvest year as well as on the severity of take-all disease in second wheat cultivars. Each of the second wheat varieties is plotted separately based on whether Cadenza (low TAB) or Hereward (high TAB) was the variety planted in the previous year in that same plot. Additionally, Hereward (high TAB), the only of the tested strains with known susceptibility to take-all, is colored orange in the charts below. Furthermore, the pattern on the bars of each cultivar corresponds to the previous year's crop, with the dots representing Cadenza being the prior year's crop and the cross hatching representing Hereward being the prior year's crop.

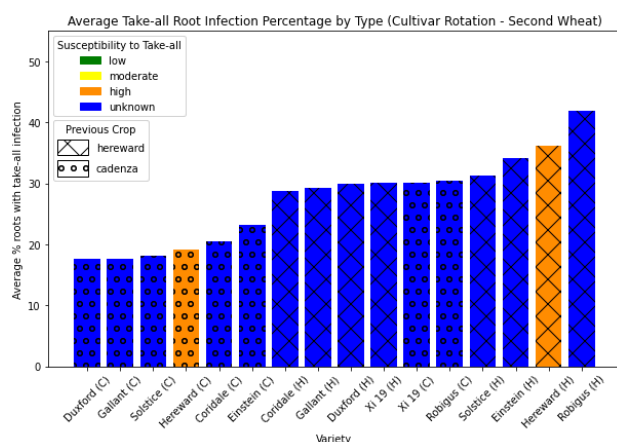


Figure 21: take-all root infection by variate with previous crop annotation

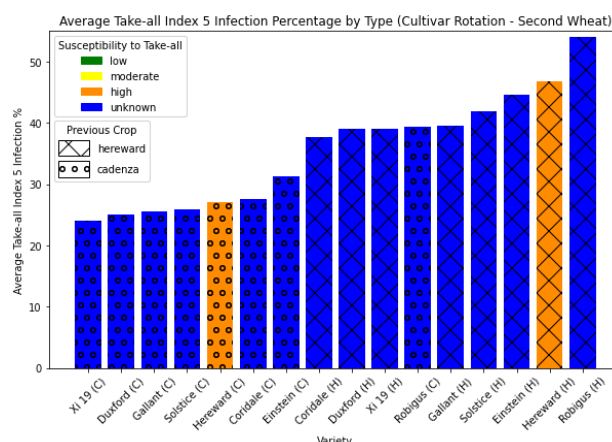


Figure 22: take-all index by variate with previous crop annotation

A visual analysis of the above bar charts shows that the use of a low TAB trait cultivar, such as Cadenza, as a first wheat crop results in a lower average percent of roots with take-all infection and lower average take-all infection index 5 infection percentages in a variety of second wheat cultivars when compared to the same second wheat cultivars planted after a high TAB trait cultivar, such as Hereward. For every variety tested in these experiments, the severity of take-all in second wheats (measured by the

take-all index 5 infection percent) of that particular variety was lower when planted after a low TAB cultivar such as Cadenza. This general sentiment is also true for the buildup of take-all in the soil of second wheat crops as for all cultivars, except Xi19, the average percent of roots with take-all infection of each variety is lower when planted after the low TAB Cadenza. Based on a visual analysis, the buildup of take-all in second wheat soil of Xi19 wheat seems to be largely unaffected by whether a low TAB or high TAB variety was grown in the same plot in the prior year.

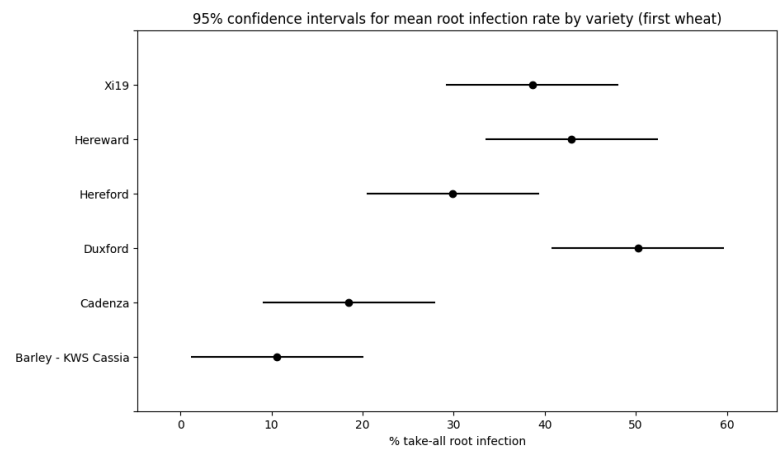
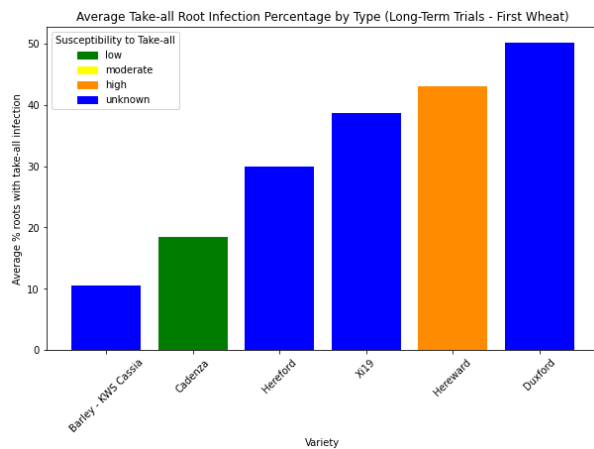
A t-test is performed to determine if the difference between the average percent of roots with take-all infection for all of the varieties when grown after Cadenza and for all of the varieties when grown after Hereward was statistically significant. The difference between the groups is determined to be statistically significant with a p-value of 0.00016. Additionally, a t-test is performed to determine if the difference between the average take-all index 5 infection percentage for all varieties when grown after Cadenza and for all varieties when grown after Hereward was statistically significant. Once again, the difference between the groups is determined to be statistically significant with a p-value of 0.000095. It follows that the use of a low TAB or high TAB trait cultivar as a first wheat crop has a statistically significant impact on the build up of take-all in the soil as well as on the severity of take-all in a variety of second wheat cultivars.

5.5.3 Long-term cultivar comparison and take-all decline

To investigate the course of a take-all epidemic from build-up in first wheat crop to take-all decline in the 4th and 5th wheat crops for various winter wheat cultivars, Rothamsted Research performed a series of long term wheat cultivar experiments in which five wheat varieties, Cadenza, Hereford, Xi19, Hereward, and Duxford, and one barley variety, KWS Cassia, were successively grown in the same plots. Data has been collected and examined for six years of the first experiment, five years of the second experiment, and four years of the third experiment. Remaining data for these three experiments as well as the final experiment is still being finalized.

An examination of the ability of different cultivars to build up take-all in soil during crop years (first wheat, second wheat, etc) is performed to determine if there is a significant difference between the average percent of roots with take-all infection found via the soil core bioassay technique on the basis of variety. The average percent of roots with take-all infection is plotted by variety for each year of the experiments, with the known low TAB cultivar, Cadenza, colored in green, and the known high TAB cultivar, Hereward, colored in orange. An example of these bar charts for the first wheat year of the first long term experiment is shown in Figure 23 below.

A Tukey test is used to determine which pairs of cultivars had a statistically significant difference in the average percent of roots with take-all infection for each year in each experiment. Only in first wheats are any pairs of cultivars determined to have a statistically significant (p-value < 0.05) difference in the average percent of roots with take-all infection across all three long-term experiments examined. The plot in Figure 24 shows the average percent of roots with take-all infection by variety for first wheats in the first long term experiment. Each variety also has a 95% confidence interval displayed on the plot. The varieties having confidence intervals that do not overlap one another on the graph are those shown by the Tukey test to be significantly different from one another in the percent of roots with take-all infection.



Figures 23 (left) and 24 (right): average take-all root infection by variate (left); 95% confidence intervals of mean root infection rate by variate (right)

This bar chart, Tukey test, and confidence interval plotting process is repeated for each harvest year of each experiment, but only showing statistically significant differences in first wheats. It follows that the type of wheat cultivar used for first wheat has a more critical impact on the buildup of take-all in the soil than the type of wheat cultivar used in any subsequent year.

An additional interest of Rothamsted Research in performing these long-term studies was to investigate the occurrence of a phenomenon known as take-all decline, in which after around four consecutive years of growing wheat in the same fields the occurrence of take-all in the wheat begins to drop off dramatically. The most obvious evidence of this occurrence appears in the six year data collected from the first experiment and is plotted in the line graph below (Figure 25).

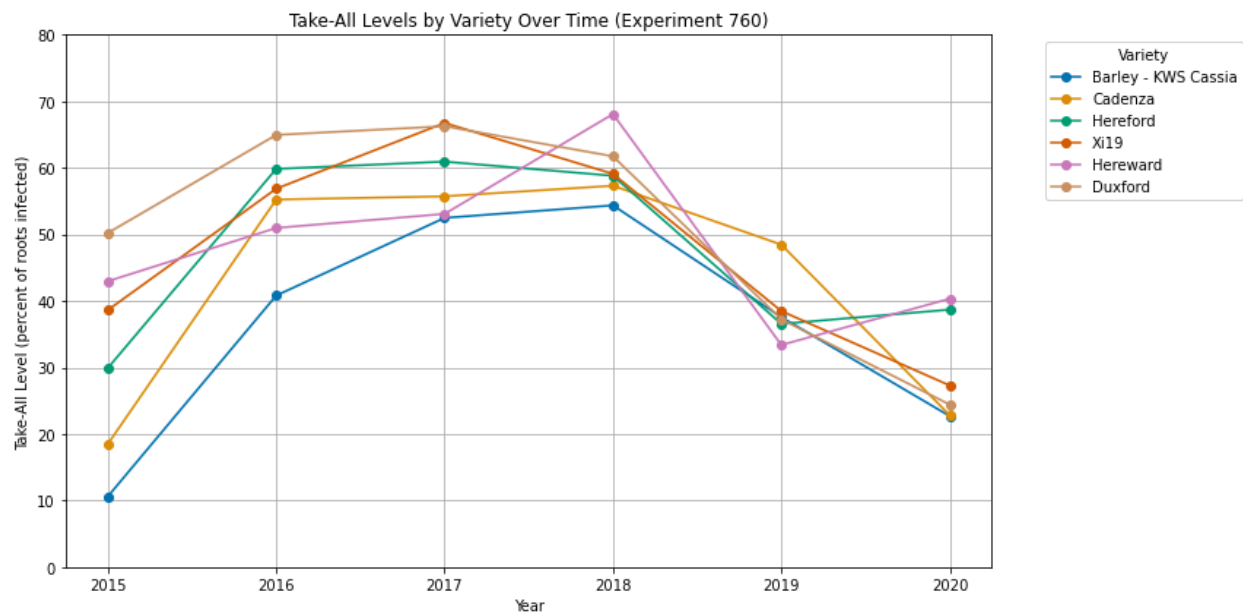


Figure 25: Take-all buildup and decline by variety over 6 years (exp. 760)

The graph plots the take-all level (by percent of roots infected) over the course of six years for five wheat varieties and one barley variety. The graph demonstrates large variability in the level of take-all in first wheats depending on the cultivar. Take-all level then increases pretty dramatically in year two for all varieties, proceeding to remain at high but more stable levels for years two to four. After year four the phenomena of take-all decline can be observed as the level of take-all across all strains generally begins to decrease. Additionally, a Tukey test is used to determine if the difference between the average take-all level for each year (averaging all strains) was statistically significant. A statistically significant difference is shown (Figure 15, Section 5.4.1) between take-all levels in year 1 and in years 2-4 as well as between take-all levels in years 2-4 and years 5-6, demonstrating the phenomenon of take-all buildup and take-all decline respectively.

An analysis of this graph raises some interesting questions for future research, particularly regarding the rate of take-all decline in Cadenza, a known low TAB trait variety. The decline in take-all level for all other varieties between years four and five seems to be more drastic than the decline between years four and five for Cadenza. This raises questions about whether low TAB cultivars have a tendency to reach take-all decline later or at a slower rate than other higher TAB cultivars. Additionally, Hereford and Hereward seem to have steady or slightly increasing levels of take-all between years five and six while all other varieties continue to decline. Further data is necessary to examine these questions in more detail.

Similar plots of take-all level over time broken down by variety for the additional long-term experiments are shown below (Figures 26 and 27), but additional data is necessary to see long term trends such as that of take-all decline. These graphs raise additional questions about the impact of soil type and field location on take-all levels as they were conducted during the same time period as the first long term study discussed above but both only reach maximum percent of roots with take-all levels beneath 50% while the first long term study had maximum take-all levels approaching 70%.

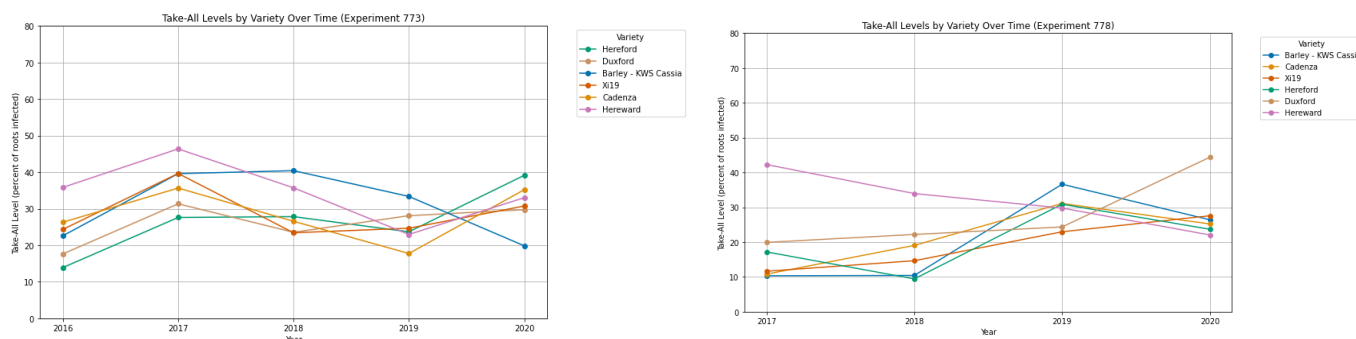


Figure 26 (left): Take-all root infection 2016 to 2020 by variate (exp. 773); Figure 27 (right): Take-all root infection 2016 to 2020 by variate (exp. 778)

Summary

Take-all disease is a significant issue in the production of wheat and determining the causes is essential to tackling food insecurity worldwide. Through years of research and experimentation, Rothamsted Research has started to identify cultivars more prone to take-all and possible factors which worsen the effects.

However, we struggle to find any strong correlation between weather factors and wheat yield. The most significant finding was that rainfall in the months around harvest were weakly correlated with root take-all infection rates, namely the spring months of May, June, and July. Other variables such as sunshine and average temperature maximums/minimums are found to be insignificant, likely due to how crop data was recorded on an annual basis. While weather is measured daily, plant measurements are taken annually, forcing data to be aggregated and models to be overfit.

Exploring relations between plant height, grain yield, straw yield, and TAB reveals questions to be explored in future research possibilities. With an R-Squared value of 0.647, the Hereward strain's straw yield could be predicted using its grain yield. It may be worthwhile exploring an inverse relationship in attempts to predict the profitability of certain strains based on their initial harvest masses. Additionally, we find little to no correlation between plant height and take-all disease in first-wheat, an unexpected result considering the literature written about such a phenomenon. It is concluded that errors in measurements and aggregation are too large and create too much variance in the data, however we speculate that other biological factors may play a role in this relationship.

Lastly, we explore the significance of wheat varieties on take-all buildup and decline. While some conclusions agree with previous studies and estimates from biologists at Rothamsted, we also find differing performances in wheat varieties past the first year. For example, while Cadenza consistently avoids high levels of take-all in first-year harvests, it also experiences a slower take-all decline than other strains in later years.

The data processing and analysis in this paper will hopefully provide useful and insightful information on take-all disease and the factors that are associated with its buildup and decline. It is hoped that future studies look to exploratory research such as this to inform experimental design decisions, biological analyses, and more in depth statistical studies.

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<https://doi.org/10.1111/j.1365-3059.2010.02375.x>

Appendix 1

Guide to Pre-Processing Tab Data and Yield Data for the Low TAB Trait Pedigree Survey

Pre step: Navigate to General/datasets/Low_TAB_trait_pedigree_survey/source_data in the sharepoint. In Visual Studio, navigate to input/Low_TAB_trait_pedigree_survey/source_data.. Inside the year folders, there are two different types of files to clean: TAB data and yield data. This is a guide for TAB data but there will be notes throughout indicating any differences in the process for yield data. There are two ipynb files, one for each of the types of files. To clean TAB data, use cleaning_tab.ipynb. To clean yield data, use cleaning_yield.ipynb.

1. Reading and Combining Data (use clean/cleaning_tab.ipynb to analyze TAB data, but similar processes for the yield files)
 - Locate values for path_tab and path_plot, each containing TAB/Wheat Cultivar data, respectively. These can be found in the same source_data folder.
 - Initialize path_tab to be the file that you wish to clean. These are examples of the file paths from 2015. Other years will be exactly the same, but with a different study name (e.g. 2016-R-CS-772)
 - Due to the programming file being in the clean folder, you must type `../` in the beginning to back out of that folder to then search for the input folder.
 - For TAB data, locate path =
`"../input/Low_TAB_trait_pedigree_survey/source_data/2015-R-CS-759/2015-R-CS-759_tab_data.txt."`
 - For yield data, locate path =
`"../input/Low_TAB_trait_pedigree_survey/source_data/2015-R-CS-759/2015-R-CS-759_yield_data.txt."`
 - Initialize path_plot to be the plot_data.csv located in the same source_data folder. (Note: for second year wheats, the plot data is the same as the previous year so 2016-R-CS-759 uses the same plot data as 2015-R-CS-759.)
 - For every type of data file, locate path_plot =
`"../input/Low_TAB_trait_pedigree_survey/source_data/2015-R-CS-759/2015-R-CS-759_plot_data.csv"`
 - Locate values for study_name and first_var, both essential for formatting.
 - The study_name will be in the first half of the path_tab file name, in a format similar to 2015-R-CS-759
 - To find first_var, open the tab_data.txt file. The top-left most value will be the first_var.
2. Creating Universal Variable Names
 - Standardize the variable names in the raw data across years as shown in the chart below, with standardized variable names on the right and the names from the raw data on the left.
 - Some of the variables in the raw files have different names depending on the study year. Make sure to check the file you are trying to clean for which variable names it uses. Then edit the code so that it recognizes the name from the raw data and changes it to the correct standardized variable name.

Notes	Initial	Standardized
	Units	plot_id
	Blocks	replicate
dependent on .txt file	Wplots/Plots	block
“ ”	Splots/col	plot
only for TAB data	%PlTa_PHarv	ww_pla_tkal_inf_pct
“ ”	%RtTa_PHarv	ww_rts_tkal_inf_pct
only for yield data	Grain85%	ww_grnyld_85pctdm_tha
“ ”	GrainDm%	ww_grn_dm_pct

3. Rename File and Export

- Locate
data.to_csv("../output/Low_Tab_Pedigree/2015-R-CS-759/2015-R-CS-759_takeall_build
up_data_agg.csv")
- Change highlighted section to the correct study name and type of file
 - Examples for plant height and yield data:
 - ../output/Low_Tab_Pedigree/2015-R-CS-759/2015_R-CS-759_ht.csv
 - ../output/Low_Tab_Pedigree/2015-R-CS-759/2015-R-CS-759_yield_data
.csv